

IFT 6268 Stats refresher

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Random variables

- ▶ The only relevant information about random variable is its distribution: (“A random variable is (i) not random, (ii) not a variable.”)
- ▶ $p(x)$ is a distribution if

$$p(x) \geq 0 \text{ and } \sum_x p(x) = 1$$

- ▶ Notational quirks:
 - ▶ p is usually heavily overloaded. The argument decides. For example, in “ $p(x, y) = p(x)p(y)$ ” each p means something different!
 - ▶ Sometimes we write X for the RV and x for the values it can take on.
 - ▶ Another common notation is $p(X = x)$.
 - ▶ $\sum_x \dots$ refers to the sum over all values that x can take on.
- ▶ For continuous x , replace $\sum$ by $\int$ (up to some measure theoretic “glitches”, that we can usually safely ignore)
- ▶ Some use the term “density” or “probability density function (pdf)” to refer to continuous $p(\cdot)$.

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Probabilities in AI

- ▶ Probabilities allow us to be explicit about uncertainties:
- ▶ Instead of representing *values*, we can “keep all options open” by defining a distribution over alternatives.
- ▶ Example: Instead of setting ' $x = 4$ ', define all of:
 $p(x = 1), p(x = 2), p(x = 3), p(x = 4), p(x = 5)$
- ▶ Benefits:
 1. Robustness (let modules tell each other their whole state of knowledge)
 2. Measure of uncertainty (“errorbars”)
 3. Multimodality (keep ambiguities around)
- ▶ We can still express ' $x = 4$ ' as a special case.

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Summarizing properties

- ▶ Any relevant properties of RVs are just properties of their distributions.
- ▶ Mean:

$$\mu = \sum_x p(x)x = E[x]$$

- ▶ Variance:

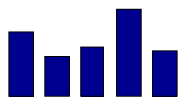
$$\sigma^2 = \sum_x p(x)(x - \mu)^2 = E[(x - \mu)^2]$$

- ▶ (Standard deviation: $\sigma = \sqrt{\sigma^2}$)

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Some useful distributions (1d)

Discrete

- ▶ **Bernoulli:** $p^x(1-p)^{1-x}$ where x is 0 or 1.
- ▶ **Discrete distribution:**  (also known as “multinoulli”)
- ▶ **Binomial, Multinomial:** Sum over Bernoulli/Discrete. (Sometimes “multinomial” is used to refer to a discrete distribution, too...)
- ▶ **Poisson:** $p(k) = \frac{\lambda^k \exp(-\lambda)}{k!}$

Continuous

- ▶ **Uniform:** 
- ▶ **Gaussian (1d):** $p(x) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$

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How to represent discrete values

- ▶ A very useful way to represent a variable that takes one out of K values:
- ▶ As a K -vector with $(K-1)$ 0's, and one 1 at position k

$$\mathbf{x} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

- ▶ This is known as **one-of- K encoding**, **one-hot encoding**, or as **orthogonal encoding**.
- ▶ Note that we can interpret $\mathbf{x}$ itself as a probability distribution.

Navigation icons

Some useful distributions (1d)

- ▶ Using a one-hot encoding allows us to write the discrete distribution compactly as

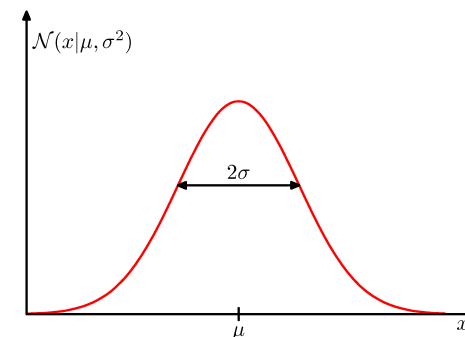
$$p(\mathbf{x}) = \prod_k \mu_k^{x_k}$$

where μ_k is the probability for state k .

- ▶ This can greatly simplify maximum likelihood calculations (see below).

Navigation icons

The Gaussian (1d)



$$p(x) = \mathcal{N}(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

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Multiple variables

- ▶ The **joint distribution** $p(x, y)$ of two variables x and y also satisfies

$$p(x, y) > 0 \text{ and } \sum_{x,y} p(x, y) = 1,$$

- Likewise, we can write

$$p(\mathbf{x}) > 0 \text{ and } \sum_{\mathbf{x}} p(\mathbf{x}) = 1,$$

for vector $\mathbf{x}$

- ▶ For discrete RVs, the joint is a *table* (or a higher-dimensional *array*).
- ▶ Everything else stays the same.



Summarizing properties, correlation

- ▶ Mean:

$$\mu = \sum_{\mathbf{x}} p(\mathbf{x}) \mathbf{x} = E[\mathbf{x}]$$

- ▶ Covariance:

$$\text{cov}(x_i, x_j) = E[(x_i - \mu_i)(x_j - \mu_j)]$$

- Covariance matrix:

$$\Sigma_{ij} = \text{cov}(x_i, x_j) \quad (\Sigma = \sum_{\mathbf{x}} p(\mathbf{x})(\mathbf{x} - \mu)(\mathbf{x} - \mu)^T)$$

- ▶ The correlation coefficient:

$$\text{corr}(X_i, X_j) = \frac{\text{cov}(X_i, X_j)}{\sqrt{\sigma_i^2 \sigma_j^2}}$$

- ▶ Two variables for which the covariance is zero are called **uncorrelated**.



Conditionals, marginals

- ▶ Everything one may want to know about a random vector can be derived from the joint distribution.

- **Marginal distributions:**

$$p(x) = \sum_y p(x, y) \quad \text{and} \quad p(y) = \sum_x p(x, y)$$

- ▶ Imagine collapsing tables.

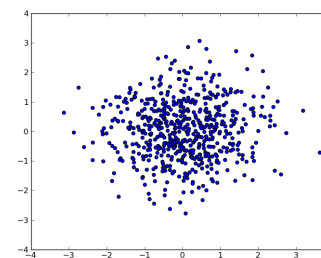
- ▶ **Conditional distributions:**

$$p(y|x) = \frac{p(x,y)}{p(x)} \quad \text{and} \quad p(x|y) = \frac{p(x,y)}{p(y)}$$

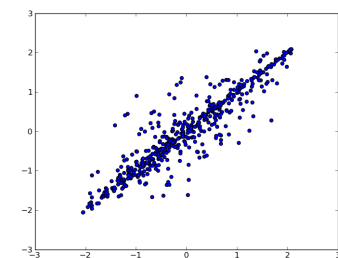
- ▶ Think of conditional as a family of distributions, “indexed” by the conditioning variable. (We could write $p(y|x)$ also as $p_x(y)$).



Correlation example



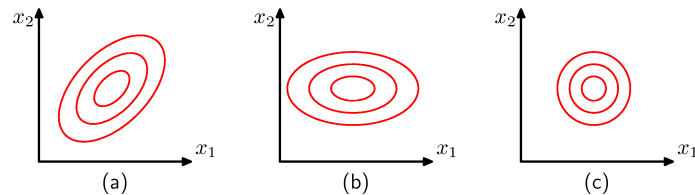
uncorrelated



correlated



The multivariate Gaussian



$$p(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{D}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$$

A fundamental formula

$$p(x|y)p(y) = p(x, y) = p(y|x)p(x)$$

- ▶ This can be generalized to more variables (“chainrule of probability”).
- ▶ A special case is **Bayes’ rule**:

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

Independence and conditional independence

- ▶ Two RVs are called **independent**, if

$$p(x, y) = p(x)p(y)$$

- ▶ Captures our intuition of “dependence”. In particular, note that this definition implies

$$p(y|x) = p(y)$$

- ▶ Independence implies uncorrelatedness, but not vice versa!
- ▶ Related: Two RVs are called **conditionally independent**, given a third variable z , if

$$p(x, y|z) = p(x|z)p(y|z)$$

- ▶ (Note that these concepts are just a property of the joint.)

Independence is useful

- ▶ Say, we have some variables, $x_1, x_2, \dots, x_K$
- ▶ Even just defining their joint (let alone doing computations with it) is hopeless for large K !
- ▶ But what if all the x_i are independent?
- ▶ Then we need to specify just K probabilities, because *the joint is the product*.
- ▶ A more sophisticated version of this idea, using *conditional independence*, is the basis for the area of *graphical models*.

Maximum likelihood

- ▶ Another useful property of independence.
- ▶ Task: Given a set of data points

$$(x_1, \dots, x_N)$$

build a model of the data-generating process.

- Approach: Fit a parametric distribution $p(x; \mathbf{w})$ with some parameters $\mathbf{w}$ to the data.
- How? Maximize the probability of “seeing” the data under your model!

Gaussian example

- ▶ What is the ML-estimate of the mean of a Gaussian?
- ▶ We need to maximize

$$L(\mu) = \sum_i \log p(x_i; \mu) = \sum_i \left(-\frac{1}{2\sigma^2}(x_i - \mu)^2 \right) - \text{const.}$$

- The derivative is:

$$\frac{\partial L(\mu)}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (x_i - \mu) = \frac{1}{\sigma^2} (\sum_i x_i - N\mu)$$

- ▶ By setting to zero, we get:

$$\mu = \frac{1}{N} \sum_i x_i$$

Maximum likelihood

- ▶ This is easy if examples are independent and identically distributed (“**iid**”):

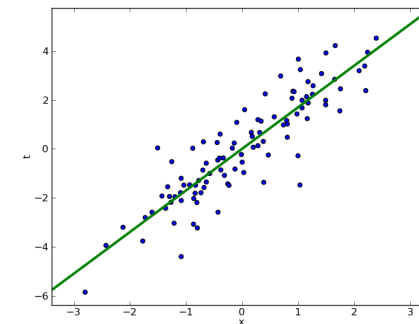
$$p(\mathbf{x}_1, \dots, \mathbf{x}_N; \mathbf{w}) = \prod_i p(\mathbf{x}_i; \mathbf{w})$$

- ▶ Instead of maximizing probability, we may maximize log-probability, because the log function is monotonic.
- ▶ So we may maximize:

$$L(\mathbf{w}) := \log \prod_i p(\mathbf{x}_i; \mathbf{w}) = \sum_i \log p(\mathbf{x}_i; \mathbf{w})$$

- ▶ Thus each example $\mathbf{x}_i$ contributes an additive component to the objective.

Linear regression



$$\mathbf{x} \rightarrow \mathbf{t}$$

- ▶ Given two *real-valued* observations $\mathbf{x}$ and $\mathbf{t}$, learn to predict $\mathbf{t}$ from $\mathbf{x}$.
- ▶ This is a *supervised learning* problem.

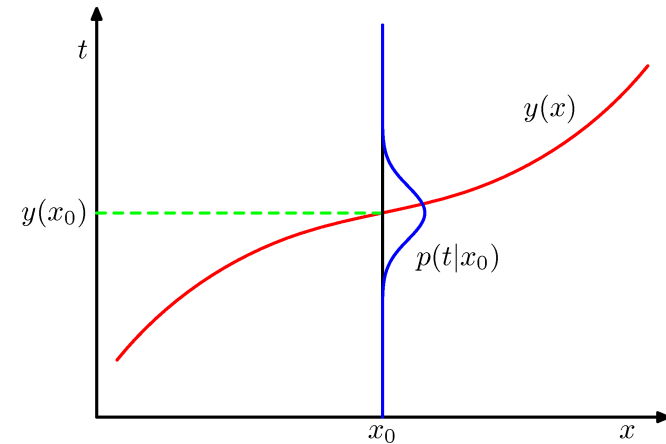
Linear regression

- ▶ We can define linear regression as a probabilistic model, if we make the following assumption:

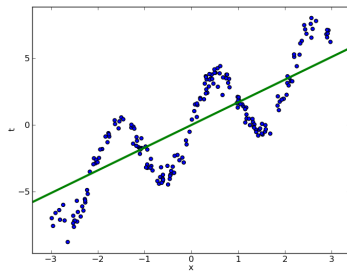
$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon$$

- ▶ In words, we assume there is a true, underlying function $y(\mathbf{x}, \mathbf{w})$, and the function values we observe are corrupted by *additive Gaussian noise*.
- ▶ Thus

$$p(t|\mathbf{x}; \mathbf{w}) = \mathcal{N}(t|y(\mathbf{x}, \mathbf{w}), \sigma^2)$$



Noise vs. dependencies we don't care about



- ▶ Actually, linear regression can work fine also with highly non-Gaussian noise.

Linear regression

- ▶ To fit the conditional Gaussian, given training data $\mathcal{D} = \{(\mathbf{x}_n, t_n)\}_{n=1}^N$, we make the iid assumption and get:

$$p(\mathcal{D}) = \prod_n \mathcal{N}(t_n | y(\mathbf{x}_n, \mathbf{w}), \sigma^2)$$

- ▶ Using monotonicity of the log, we may again maximize the log-probability (or minimize its negation):

$$\text{minimize} \quad \sum_{n=1}^N (t_n - \mathbf{w}^T \mathbf{x}_n)^2 + \text{const.}$$

Least squares

- ▶ To optimize with respect to $\mathbf{w}$, we differentiate:

$$\frac{\partial E}{\partial \mathbf{w}} = - \sum_{n=1}^N (t_n - \mathbf{w}^T \mathbf{x}_n) \mathbf{x}_n^T$$

- ▶ Setting the derivative to zero:

$$0 = - \sum_{n=1}^N t_n \mathbf{x}_n^T + \mathbf{w}^T \left(\sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^T \right)$$

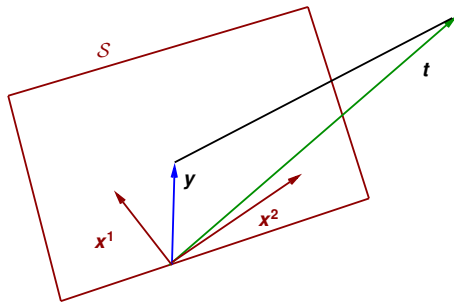
yields the solution

$$\mathbf{w} = \left(\sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^T \right)^{-1} \left(\sum_{n=1}^N t_n \mathbf{x}_n^T \right)$$

- ▶ (It can be instructional to write down the case for 1-d inputs, if this confuses you)



Geometrical interpretation



- ▶ We can think of the squared error $\sum_n (t_n - \mathbf{w}^T \mathbf{x}_n)^2$ as the squared norm of the difference between two N -dimensional vectors $\mathbf{t}$ and $\mathbf{y}$, containing all training-outputs and model-predictions.
- ▶ The vector $\mathbf{y} (= \mathbf{X}\mathbf{w})$ has to lie in the space $\mathcal{S}$ spanned by the *columns* $\mathbf{x}^i$ of $\mathbf{X}$.
- ▶ It is exactly the *orthogonal projection* of $\mathbf{t}$ onto $\mathcal{S}$



Least squares

- ▶ We can write this more compactly with the following definitions:

$$\mathbf{t} = \begin{pmatrix} t_1 \\ \vdots \\ t_N \end{pmatrix}$$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_N \end{pmatrix}$$

This allows us to write the the solution as

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{t}$$

“The normal equations”.



Linear classification

$$\mathbf{x} \rightarrow t$$

- ▶ A prediction task, where the outputs, t , are discrete (that is, they can take on one of K values, $c_1, \dots, c_K$), the task is a *classification task*.
- ▶ Like linear regression, this is a *supervised learning* problem.



(Multi-class) logistic regression

- Logistic regression defines a probabilistic model over classes given inputs as follows:

$$p(c_k | \mathbf{x}) = \frac{\exp(\mathbf{w}_k^T \mathbf{x}_n)}{\sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{x}_n)}$$

where $\mathbf{w}_1, \dots, \mathbf{w}_K$ are parameters.

- The exp-function ensures positivity, and the normalization that the outputs sum to one.
- (In practice, one usually adds constant “bias”-terms inside the exp’s)

Navigation icons

Multi-class logistic regression

- Represent discrete one-hot labels row-wise in a matrix $\mathbf{T}$, like we did before for continuous vectors.
- The negative log-likelihood cost, assuming iid training data, can then be written

$$\begin{aligned} E(\mathbf{W}; \mathcal{D}) &= -\log \prod_n p(\mathbf{t}_n | \mathbf{x}_n) \\ &= -\log \prod_n \prod_k p(c_k | \mathbf{x}_n)^{t_{nk}} \\ &= -\sum_n \sum_k t_{nk} \log p(c_k | \mathbf{x}_n) \\ &= -\sum_n \sum_k t_{nk} (\mathbf{w}_k^T \mathbf{x}_n - \log \sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{x}_n)) \\ &= -\sum_n (\mathbf{w}_{t_n}^T \mathbf{x}_n - \log \sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{x}_n)) \end{aligned}$$

Navigation icons

Multi-class logistic regression

- In contrast to linear regression, there is no closed-form solution for $\mathbf{W}$.
- But one can use gradient-based optimization to minimize $E(\mathbf{W}; \mathcal{D})$ iteratively.
- The gradient with respect to each parameter-vector $\mathbf{w}_k$ is

$$\begin{aligned} \frac{\partial E(\mathbf{W}; \mathcal{D})}{\partial \mathbf{w}_k} &= -\sum_n t_{nk} \mathbf{x}_n - \frac{\exp(\mathbf{w}_k^T \mathbf{x}_n)}{\sum_j \exp(\mathbf{w}_j^T \mathbf{x}_n)} \mathbf{x}_n \\ &= \sum_n (p(c_k | \mathbf{x}_n) - t_{nk}) \mathbf{x}_n \end{aligned}$$

- It can be shown that $E(\mathbf{W}; \mathcal{D})$ is convex, so there are no local minima.

Navigation icons

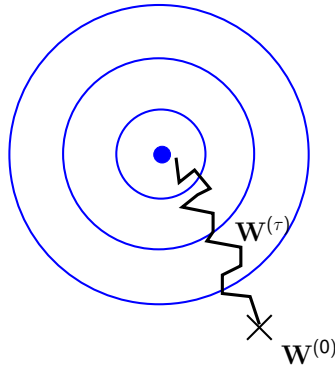
Learning with stochastic gradient descent

$$\mathbf{W}^{(\tau+1)} = \mathbf{W}^{(\tau)} - \eta \frac{\partial E_n}{\partial \mathbf{W}}$$

- Here, τ denotes the iteration number, and E_n is the cost contributed by the n^{th} training case (one term in the sum over n).
- Parameters are initialized to some random starting value $\mathbf{W}^{(0)}$.
- η is called **learning rate**, and it is typically set to a small real value (such as, $\eta = 0.001$). It may be reduced as learning progresses.
- It is convenient to think of $\mathbf{W}$ as a vector not a matrix when doing learning. (Think of it in “vectorized” form: $\text{vec}(\mathbf{W})$)

Navigation icons

Learning with stochastic gradient descent



- ▶ Since the algorithm visits one training-case at a time, it will jitter around an idealized “average path” towards the optimum.
- ▶ That’s why it’s called “stochastic” gradient descent.
- ▶ One could use the gradient of the whole sum instead but that is often slower (because of redundancies in the data).

Random variables and information

- ▶ "Probabilities allow us to be explicit about uncertainty". So how can we *measure* uncertainty?
- ▶ Idea: Define the *information content*

$$\log\left(\frac{1}{p(x)}\right) = -\log(p(x))$$

contained in a random event.

- ▶ The information content is additive for independent events.
- ▶ So if we use $\log_2$ and fair coin tosses, then the information content is measured in *bits* and it exactly fits our intuition.

The “logsumexp”-trick

- ▶ Expressions like

$$\frac{\exp(\mathbf{w}_k^T \mathbf{x}_n)}{\sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{x}_n)}$$

are very common but *highly unstable*, because the “exp” in the denominator can cause an under- or overflow.

- ▶ Never compute sums $\sum_i \exp(a_i)$ naively.
- ▶ Add a constant A to each argument in all exp's, so that even the largest argument is small; then undo the operation after computing the sum!
- ▶ Many software packages supply a convenience function “logsumexp” for this purpose:

logsumexp

$$\text{logsumexp}(\mathbf{a}_1, \dots, \mathbf{a}_K) = \log \left(\sum_i \exp(\mathbf{a}_i + A) \right) - A$$

with $A = -(\max_j \mathbf{a}_j)$

Entropy

- ▶ To measure uncertainty, we define the entropy

$$H(X) = - \sum_x p(x) \log p(x)$$

which is the average information content.

- ▶ For continuous RV:

$$H(X) = - \int_x p(x) \log p(x) dx$$

- ▶ The more uniform, the more uncertain. The more “peaky”, the more certain.
- ▶ Question: Which probability distribution has maximum entropy, given mean and (co)variance(s)?

KL divergence

- ▶ Closely related to entropy is the Kullback-Leibler divergence (KL divergence) (or “relative entropy”):

$$\text{KL}(p||q) = \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right)$$

- ▶ The KL divergence measures the dissimilarity between two distributions.
- ▶ It is not symmetric.

The Frequentist – Bayesian debate

- ▶ Probability theory tells us how to *calculate* with probabilities.
- ▶ As scientists, we may ask how to *interpret* a probabilistic expression, like $p(x = 1) = 0.7$
- ▶ There are two common interpretations:
 1. *Frequentist*: “The *relative frequency* of x in a (possibly infinite) population of trials is 0.7”
 2. *Bayesian*: “*I believe* that x is 1 with certainty 0.7”
- ▶ The Bayesian view used to be contentious because it is less intuitive.
- ▶ But it gives us the freedom to turn model parameters into random variables.

Mutual Information

- ▶ The mutual information between two random variables x, y with joint density $p(x, y)$ is defined as

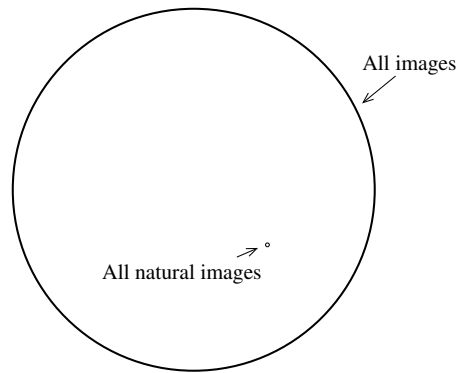
$$\text{MI}(x, y) = \sum_{x, y} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right)$$

- ▶ It is the KL divergence between $p(x, y)$ and the joint of two perfectly independent random variables (with marginals $p(x)$ and $p(y)$).
- ▶ Thus, MI measures the *dependence* between x and y .
- ▶ It is nonnegative, and it is zero iff x and y are independent, in other words iff $p(x, y) = p(x)p(y)$.

Reading

- ▶ A good introduction to most of the concepts discussed in this class can be found in:
Pattern Recognition and Machine Learning. C. Bishop.
Springer, 2006.
- ▶ Most drawings were taken from that book.

Natural images are not random



- ▶ Constraining the capacity of **s** forces learner to “zoom” into where the data is.